

Python Modules Reference — 1-Page Note

1. What Are Modules?

A module in Python is a file containing Python code (variables, functions, classes). They help in code reuse, organization, and maintainability.

Importing a module:

```
import module_name
```

```
from module_name import item
```

```
import module_name as alias
```

2. Built-in Standard Library Modules

OS & System Modules

Module	Purpose	Example
os	Interact with operating system	os.listdir()
sys	Command-line args & interpreter info	sys.argv
subprocess	Run external commands	subprocess.run()
platform	System details	platform.system()

Math & Random

Module	For	Example
math	advanced math	math.sqrt(16)
statistics	mean, median	statistics.mean([1,2])
random	random numbers	random.randint(1,10)

Date & Time

Module	For	Example
datetime	date/time handling	datetime.datetime.now()

Module	For	Example
time	delays, timestamps	time.sleep(1)

File, Data & JSON

Module	Purpose	Example
json	JSON read/write	json.loads()
csv	CSV handling	csv.reader()
shutil	file operations	shutil.copy()
pathlib	object-oriented file paths	Path('file.txt')

Networking & Web

Module	Purpose
requests*	HTTP requests (external pkg)
urllib	URL handling
socket	Networking programming

Data Structures

Module	For
collections	deque, counter, namedtuple
heapq	heap/priority queue
array	typed arrays

Regex & Parsing

Module	For
re	regular expressions

Module	For
html	HTML parsing/escaping
xml.etree.ElementTree	XML parsing

3. Creating Your Own Module

file: mymath.py

```
def add(a, b):
```

```
    return a + b
```

import it:

```
import mymath
```

```
print(mymath.add(2, 3))
```

4. Installing External Modules

Use pip:

```
pip install requests
```

```
pip install flask
```

Check installed packages:

```
pip list
```

5. Useful “Must-Know” Modules for Full-Stack Python

- Flask / Django → backend web framework
- FastAPI → next-gen APIs
- SQLAlchemy → database ORM
- Pillow → image processing
- NumPy / Pandas → data handling
- pytest / unittest → testing
- dotenv → load environment variables

